Detecting and quantifying wash trading of non-fungible tokens

Jonathan Chow

Detecting and quantifying wash trading of non-fungible tokens

In contrast to fungible tokens, where each token is identical and can be exchanged on a one-to-one basis, non-fungible tokens are unique and indivisible digital assets.



Detecting and quantifying wash trading of non-fungible tokens

Wash trading is a process where a trader buys and sells a security for the express purpose of feeding misleading information to the market. (Chen & Anderson, n.d.)

Motivation

12B total wash trading on Terraforms

- 4.6B wash trading done by just 2 wallets

Wash trading made up 23.4% of unadjusted trading volume across all marketplaces

- Wash trading is more prevalent on NFT marketplaces that provide token incentives for NFT trades, where wash trading amounted up to 85% of unadjusted trading volume on X2Y2 in the same month.

Metrics such as trading volume and floor price are typical indicators for increasing a collection's ranking in an NFT marketplace

Collection Name	Actual Volume	Wash Trading Volume	Total Volume	Wash Trading %
terraforms-by-mathcastles	22,199,051.52	12,187,594,937.77	12,209,793,989.28	99.82%
meebits	423,676,689.36	9,454,611,900.84	9,878,288,590.19	95.71%
dotdotdots	23,598,187.62	2,684,871,287.91	2,708,469,475.52	99.13%
more-loot	1,197,669.91	1,218,451,779.58	1,219,649,449.49	99.9%
dreadfulz	4,715,720.22	1,144,087,037.4	1,148,802,757.62	99.59%
loot-for-adventurers	267,631,533.24	738,234,934.63	1,005,866,467.87	73.39%
audioglyphs	4,459,311.23	378,755,710.11	383,215,021.34	98.84%
catgirl-academia	18,996.25	339,540,315.9	339,559,312.15	99.99%
cryptophunksv2	4,101,653.84	276,676,297.07	280,777,950.91	98.54%
hashmasks	95,533,368.05	233,001,040.66	328,534,408.71	70.92%

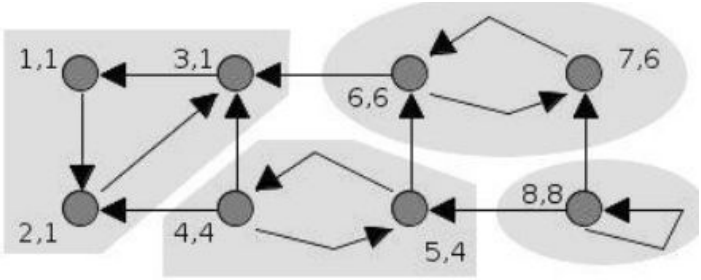
Buyer Address	Total Wash Trading Volume
0xa53496b67eec749ac41b4666d63228a0fb0409cf	2,381,770,633.53
0xd73e0def01246b650d8a367a4b209be59c8be8ab	2,377,841,614.74
0x718a085f6d45a955395881bceff30ec97ffb1013	1,008,671,982.2
0xf6764f702650eb66b2bd7368e2fef7133a8f4f80	970,299,962.67
0x32b8bfad45b330b81899df23ff6aac609781ce56	766,657,856.17

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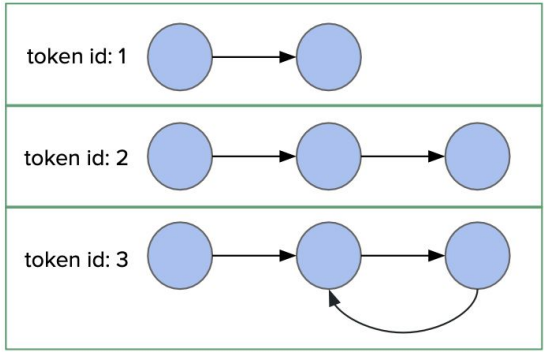
Background

Background (SCC Detection)

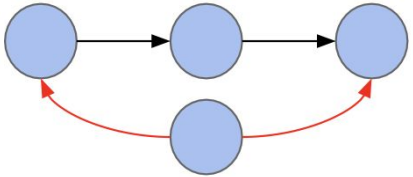
- **Tarjan's Algorithm:** Most prevalent detection method is the use of identifying strongly connected components in the NFT's asset transfer graph.
  - A directed graph is strongly connected if there is a directed path from each vertex to every other vertex in the graph.
  - In a particular study, when more than five cycles in an SCC are detected, all trades in the SCC are considered wash trades.
- Asset Transfer graph
- Asset Trade (Sales) graph
- User Transaction graph (Multiple types of transaction)
  - Only using ETH (Ethereum native token) transaction in this study
- **Profits Tracking:** Tracking profits to see value derived from a wash trade



Strongly Connected Components



Asset Transfer OR Asset Trade (Sales) graph



User Transaction Graph

Background (Open source research)

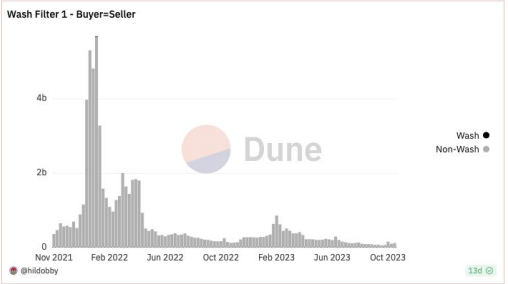
Open source research such as Dune and Footprint Analytics are typically created by for-profit organisations, but are well-cited in the industry

- Buyer = Seller
- Back and Forth Trades
- Bought NFT 3 times
- Buyer and Seller 1st funded by same wallet
- Flash Loan Transactions

Wash Filter 1 - Buyer=Seller

This filter aims to remove any trades where the buyer's address is the same as the seller's (yes, it's possible). It happens on most Marketplaces you know as a wash trading method, here are a few examples: on OpenSea, on LooksRare, on X2Y2 & on Blur.

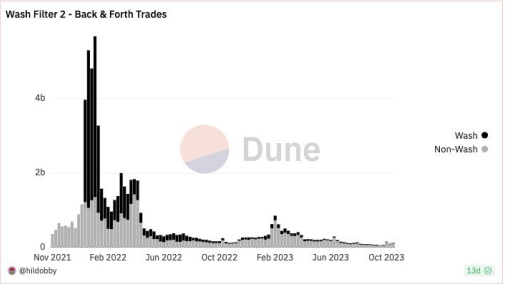
Below you can see the resulting USD trading volume with this filter applied:



Wash Filter 2 - Back & Forth Trades

Some NFTs are being traded back and forth between two wallets. Whenever a matching trade with seller and buyer inverted is detected, the trade is flagged as a wash trade.

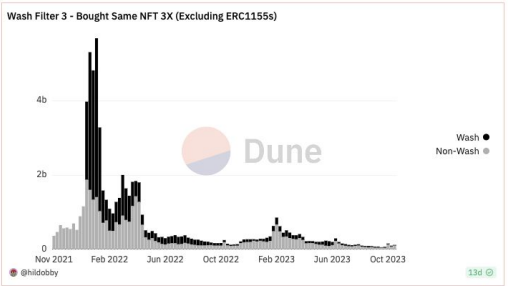
Below you can see the resulting USD trading volume with this filter applied:



Wash Filter 3 - Bought NFTs 3X (Excluding ERC1155s)

Here, if the same address appears to have bought the same NFT 3X, those transactions will be labelled as wash trades. Due to their semi-fungibility, ERC1155s are exempted from this filter.

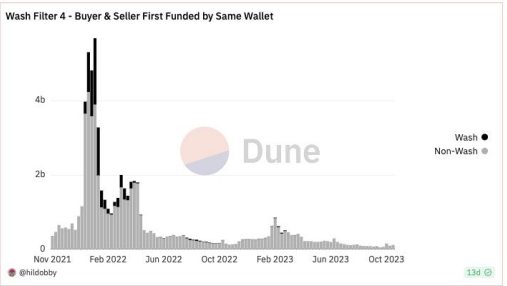
Below you can see the resulting USD trading volume with this filter applied:



Wash Filter 4 - Buyer & Seller First Funded by Same Wallet

This looks up the first ever address that funded the seller and the buyer's address. If those match, it is flagged as a wash trade, same happens if the buyer first funded the seller or vice versa. Traders first funded by any of these ~500 CEX address or a tornado smart contract are exempted from this filter)

Below you can see the resulting USD trading volume with this filter applied:

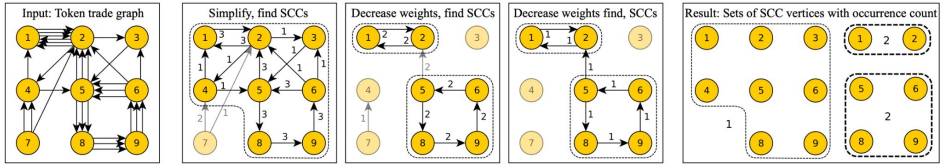


Wash Filter 5 - Flash Loan Transactions

This filter checks if a flash loan was made in the same transaction as the trade, if that's the case, then the trade is flagged as a wash trade.

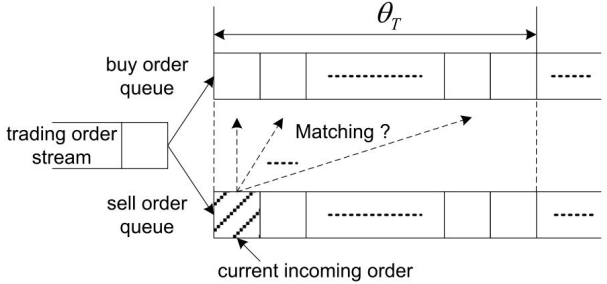
Below you can see the resulting USD trading volume with this filter applied:

Background (DEX and TradFi)

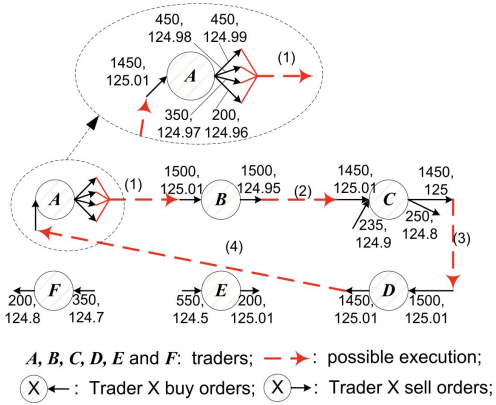


- **SCC Detection:** A user transfer graph of the fungible token as input, and get series of wash trades as a result. In several studies, unlike NFT's SCC detection, the graph has multiple tokens.
- **Submission Interval Matching:** Typically, wash trades have tighter submission intervals on matching orders to minimise the risk of the order being unintentionally picked up by another individual.
- **Volume Matching:** volume matching as a methodology does not work in NFT wash trading detection due to the non-fungibility of NFTs.

SCC Detection



Submission Interval Matching



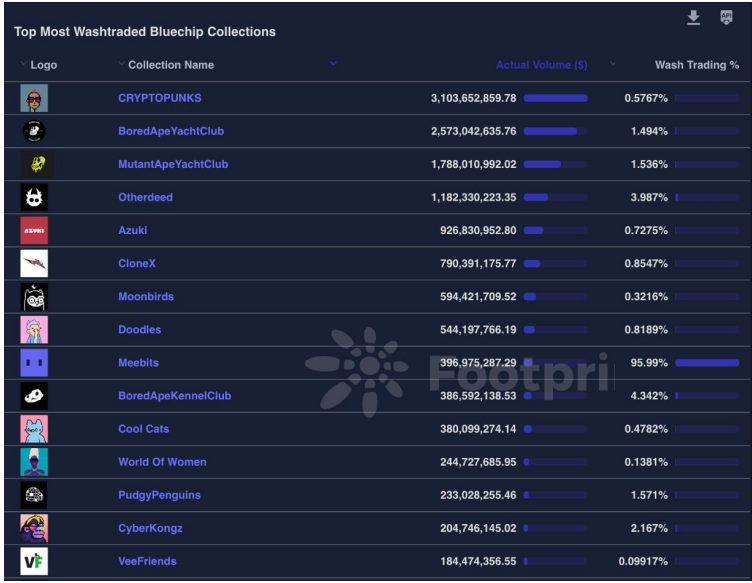
Volume Matching

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Proposed Approach

Challenges with existing approach

- SCC used alone on an asset transfer graph is bypassed by users easily due to the anonymous nature of a blockchain, where users typically create multiple accounts to perform wash trading. The SCC detection method requires a user to reuse an account one or more times, to detect the wash trading of that asset.
- Detection is largely detecting users who do not spend time masking their trades, which many still do not do as NFTMs have yet to enforce consequences to wash trading
- Trend of higher marketcaps having the same or smaller wash trading percentages is indicative of some hidden wash trades.



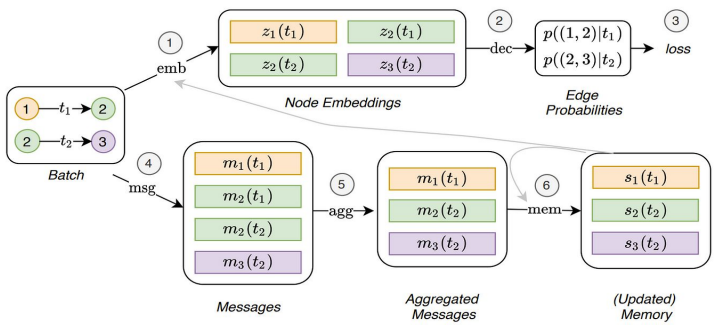
Top Most Washtraded Bluechip Collections

Logo	Collection Name	Actual Volume ($)	Wash Trading %
	CRYPTOPUNKS	3,103,652,859.78	0.5767%
	BoredApeYachtClub	2,573,042,635.76	1.494%
	MutantApeYachtClub	1,788,010,992.02	1.536%
	Otherdeed	1,182,330,223.35	3.987%
	Azuki	926,830,952.80	0.7275%
	CloneX	790,391,175.77	0.8547%
	Moonbirds	594,421,709.52	0.3216%
	Doodles	544,197,766.19	0.8189%
	Meebits	396,975,287.29	95.99%
	BoredApeKennelClub	386,592,138.53	4.342%
	Cool Cats	380,099,274.14	0.4782%
	World Of Women	244,727,685.95	0.1381%
	PudgyPenguins	233,028,255.46	1.571%
	CyberKongz	204,746,145.02	2.167%
	VeeFriends	184,474,356.55	0.09917%

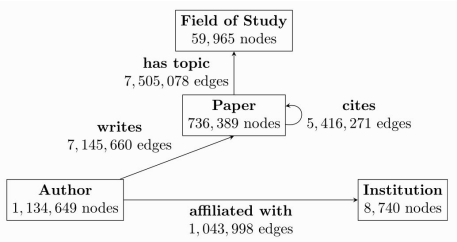
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Proposed Approach

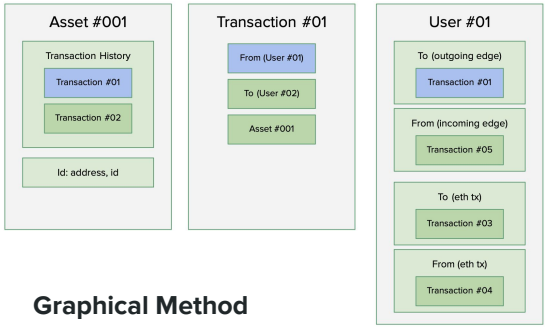
- **Graphical Neural Networks (Initial plan)**
  - **TGNs:** A time labelled message passing model is appropriate for a series of block labelled transactions
  - **Heterogeneous graph networks:** Attempting to detect structures in the data
- **Graphical Method:** Improving on current SCC implementations by using a ETH-based (Ethereum native token) user transaction graph in coordination with an asset transfer graph



Temporal Graph Networks: Temporal Analysis



Heterogeneous graph networks: Feature Engineering



Graphical Method

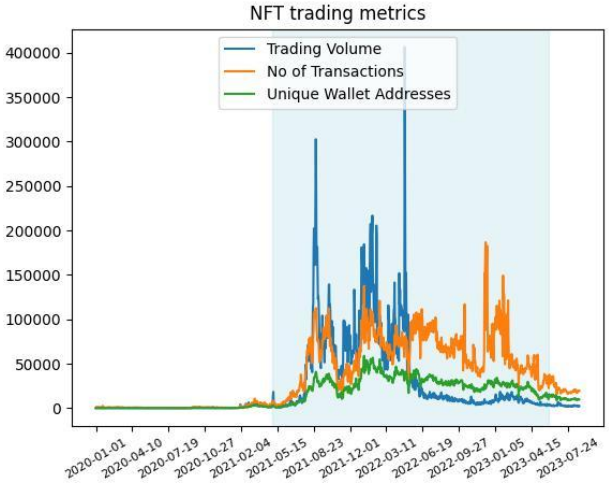
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Detailed Steps

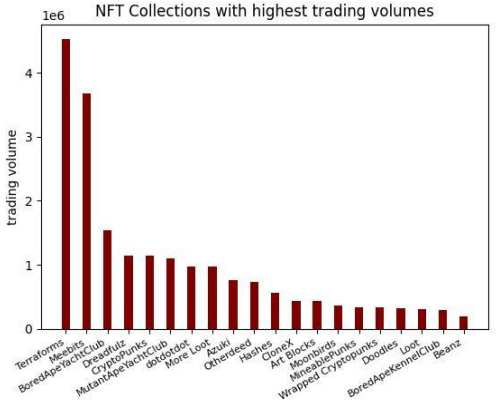
Step 1: Data Selection

- **Time Period**
  - **Selected:** 1 May 2021 to 1 June 2023
    - 40.6M in daily trading volume and 60,351 in daily transactions, compared to 0.5M and 1,332 transactions prior
- **NFT Collections:** top 20 ERC-721 NFT collections, rank ordering them by trading volume

Terraforms, Meebits, BoredApeYachtClub, Dreadfulz, CryptoPunks, MutantApeYachtClub, dotdotdot, More Loot, Azuki, Otherdeed, Hashes, CloneX, Art Blocks, Moonbirds, MineablePunks, Wrapped Cryptopunks, Doodles, Loot, BoredApeKennelClub, Beanz



NFT trading metrics between 2020 and 2023



Top 20 NFT collections by trading volume

Step 1: Data Selection

- **API Providers**
  - **Moralis API:** Time series transaction data
  - **Footprint Analytics API:** Aggregated data metrics
- **Datasets**
  - **Dataset 1** is a dataset of all **NFT Trades** on the Open Sea NFT Marketplace for the top 20 NFT collections by trading volume
  - **Dataset 2** is a dataset of all **NFT Transfers** for the top 20 NFT collections by trading volume
  - **Dataset 3** is a dataset of all **ETH Transfers** to and from all wallet addresses that participated in either an NFT trade or transfer of the Terraform NFT collection (No. 1 collection by trading volume during the time period)
  - **Dataset 4** is a list of wallet **addresses suspected** of wash trading, created by Footprint Analytics
  - **Dataset 5** is a **list of wash trades**, created by Footprint Analytics
  - **Dataset 6** is a **list of the top 100 collections** by volume across all time, with its trading volume, and wash trade percentage during the time period, created by Footprint Analytics

No.	Dataset	No. of Rows	Details
1	NFT Trades	547,361	Transaction Details
2	NFT Transfers	2,686,773	Transaction Details
3	ETH Transfers	9,299,772	Transaction Details
4	FA Suspicious Addresses	104,857	address, trading_value, trades
5	FA Tainted Tx	104,857	tx_hash
6	FA WashTrade Ratios	100	total_volume, wash_trading_volume, wash_trading_percentage

List of Datasets

Collection	NFT Trades	NFT Transfers	Collection	NFT Trades	NFT Transfers
Terraform	13,152	55,141	Hashes	1,071	5,855
Meebits	21,466	76,900	CloneX	23,006	134,551
BoredApeYachtClub	19,777	199,283	Art Blocks	199,165	518,174
Dreadfulz	13,341	43,506	Moonbirds	11,476	71,690
CryptoPunks	0	27,242	MineablePunks	3,505	10,530
MutantApeYachtClub	39,321	271,260	Wrapped Cryptopunks	18	20,153
dotdotdot	10,380	20,495	Doodles	26,255	118,746
More Loot	6,190	153,297	Loot	12,442	34,361
Azuki	28,898	131,332	BoredApeKennelClub	25,515	113,714
Otherdeed	64,273	506,076	Beanz	28,110	174,467

Breakdown of Datasets 1 and 2 by collection

Step 2: Data Preprocessing

- **Required Data:** address, token id, seller, buyer, block number, transaction hash (the rest are backfilled if not present)
- **Transfers**: (1) “token_ids” (2) seller and buyer addresses (3) the price token address (4) price value decimals (5) the block number hex
    - (1) the field “token_ids” is filtered and processed for multiple ids during the preprocessing stage, (2) seller and buyer addresses are also used as from and to addresses respectively, (3) the price token address is left as an empty string for the native Ethereum token (ETH), (4) the price value doesn’t record the decimals as part of the dataset, and (5) the block number is converted from a hexadecimal to a decimal value during preprocessing.
- **Trades**: Order fulfilled as Offer[] and Consideration[]
- **ETH Transfers:** ETH transfers dataset has been reduced to the transfers of all wallet addresses that participated in either an NFT trade or transfer of the Terraform NFT collection during the aforementioned time period.
    - 323M to 9.3M data points
    - filtering out for zero value ETH transfers
    - creates a Transaction Details object

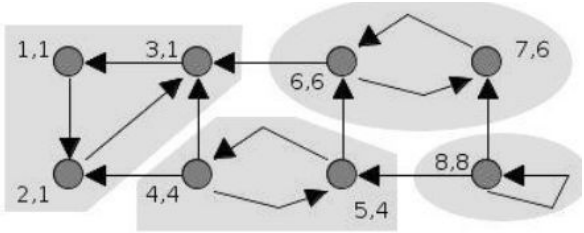
```
TransactionDetails = {
  address: '0x4e1f41613c9084fdb9e34e11fae9412427480e56',
  token_ids: 6493,
  seller_address: '0x1bcb63c4dc7ea87561faf8dfa271f4a6ee60bcc1',
  buyer_address: '0x69efd597a3f7018ce35f38f53ba05d42c4f6b1c3',
  marketplace_address: '0x00000000000000adc04c56bf30ac9d3c0aaf14dc',
  price_token_address: '',
  price: 1990000000000000000,
  block_number: 17321906,
  tx_hash:
'0x61ef0a728ab2091d99554f9cfe0c39d4dd403616df0932fcb5914025bb7d4857'
}
```

Transaction Details

No.	Dataset	No. of Rows	Details
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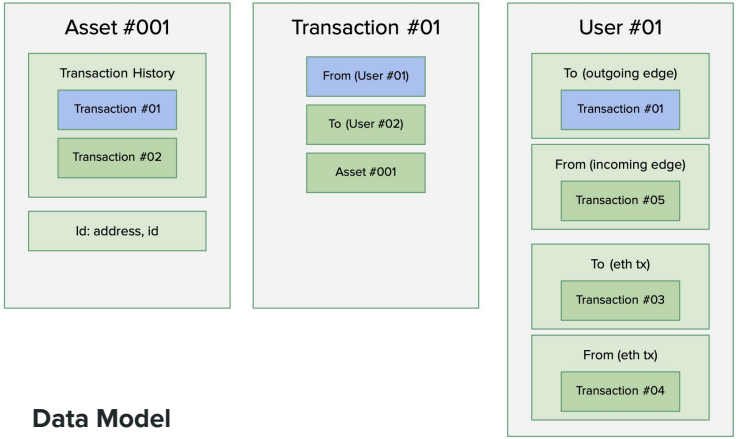
List of Datasets

Step 3: SCC Detection



Strongly connected components

- **Create Data Model / Graph**
  - **Transaction Details:** Creates Transaction object, which creates a sender, receiver, and asset
  - **Graphs**: Asset transfer graph and a user transaction graph
- **Create Asset transfer graph**
  - Sort transaction history, then for each transaction, create 2 nodes
  - Missing data: if the previous receiver is not the current sender



Data Model

Step 3: SCC Detection

Algorithm

- For a random node, DFS, if there is a back edge, save new low-link value
- For all same low-link value indicate SCC
- Start at a new random node
- Input: Directed graph
- Output: Set of SCCs, or sets of vertices

SCC Detection

- **2 directed cycles**: count the number of “back-edges” by comparing the low-link value to the index value
- **number of wash trades**: for all SCCs flagged, or have at least 2 cycles, every edge in the SCC counts as a single wash trades

```
algorithm tarjan is
    input: graph G = (V, E)
    output: set of strongly connected components (sets of vertices)

    index := 0
    S := empty stack
    for each v in V do
        if v.index is undefined then
            strongconnect(v)

    function strongconnect(v)
        // Set the depth index for v to the smallest unused index
        v.index := index
        v.lowlink := index
        index := index + 1
        S.push(v)
        v.onStack := true

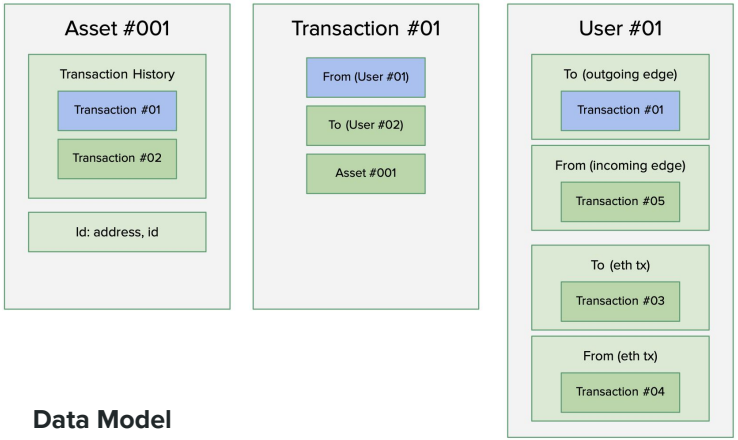
        // Consider successors of v
        for each (v, w) in E do
            if w.index is undefined then
                // Successor w has not yet been visited; recurse on it
                strongconnect(w)
                v.lowlink := min(v.lowlink, w.lowlink)
            else if w.onStack then
                // Successor w is in stack S and hence in the current SCC
                // If w is not on stack, then (v, w) is an edge pointing to an SCC already found and must be ignored
                // The next line may look odd - but is correct.
                // It says w.index not w.lowlink; that is deliberate and from the original paper
                v.lowlink := min(v.lowlink, w.index)

        // If v is a root node, pop the stack and generate an SCC
        if v.lowlink = v.index then
            start a new strongly connected component
            repeat
                w := S.pop()
                w.onStack := false
                add w to current strongly connected component
            while w ≠ v
            output the current strongly connected component
```

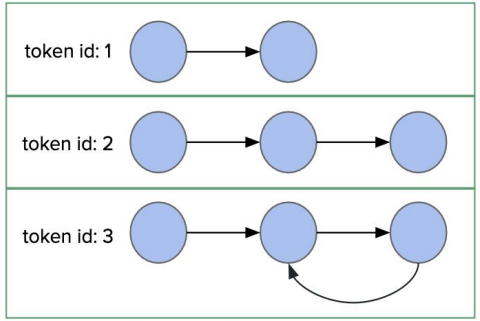
Tarjan’s Algorithm

Step 4: Wash Trading Detection

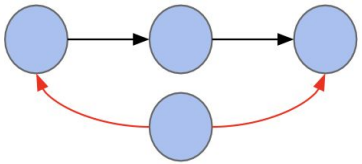
- **3 types of wash trading detection algorithms**
  - **SCC A:** SCC detection on an asset transfer graph
  - **SCC AU:** SCC detection on an asset transfer graph and user transaction graph
  - **SCC AC:** SCC detection on an asset transfer graph and simple cycle detection on an undirected user transaction graph
- **SCC AU:** SCC on both graphs, DFS max depth 30 for ETH-only branches, only SCC with asset transfer edge counts to wash trade
- **SCC AC:** SCC on Asset transfer graph, BFS to depth 2, counting all edges as undirected. Append assets transfers / trades to wash trades if cycle detected (and not already detect by SCC)



Data Model



Asset Transfer OR Asset Trade (Sales) graph



User Transaction Graph

Detecting and quantifying wash trading of non-fungible tokens

Evaluation

Evaluation: Method SCC A

- **Missing data:** amounted to up to 3% in the NFT trades dataset (dataset 1), and amounted up to 0.5% in the NFT transfers dataset (dataset 2). Found on asset transfer graph creation
- **Dataset 1:** much lower detection percentage
  - Dataset 1 is subset on Dataset 2
  - Melding data causes missing data percentage to 65% from 0.5%
  - Perhaps address translation error due to proxies or internal transactions might have been the cause of the confusion.
- **Dataset 2:** Cosine Similarity of 0.63 vs 0.82 (random fluctuations in data)
  - FA detection not revealed, might not use SCC or doesn't require 2 cycle

Collection	Wash Trade Share (Dataset 1)	Wash Trade Share (Dataset 2)	FA's Wash Trade Share
Terraform	8.37	45.89	99.76
Meebits	7.82	34.51	96.02
BoredApeYachtClub	1.01	9.78	1.46
Dreadfulz	8.30	78.97	98.45
CryptoPunks	0.00	0.05	0.61
MutantApeYachtClub	2.75	6.17	1.46
dotdotdot	7.02	35.26	99.12
More Loot	0.37	28.27	99.50
Azuki	5.22	21.05	0.54
Otherdeed	2.80	3.41	3.95
Hashes	9.17	58.27	NA
CloneX	1.17	0.27	0.81
Art Blocks	15.37	59.03	15.58
Moonbirds	9.90	53.41	NA
MineablePunks	10.70	58.27	59.56
Wrapped Cryptopunks	6.21	9.92	NA
Doodles	8.61	10.4	0.82
Loot	15.55	96.03	73.43
BoredApeKennelClub	5.58	7.73	4.38
Beanz	13.05	17.33	2.52

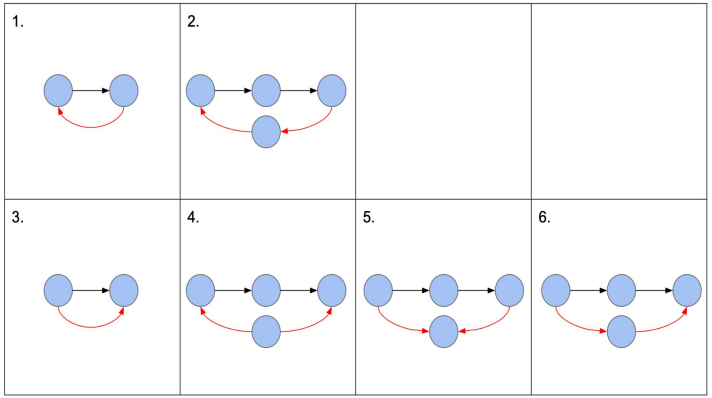
Wash Trade Share of all collections using method SCC A

Evaluation: Method SCC AU / AC

- **SCC A < SCC AU < SCC AC:** for dataset 2
- **All 2 depth cycles:** black arrows vs red arrows
  - Depth 2 selection
  - Dataset 3: user transaction graph in this study is only the transfers of all wallet addresses that participated in either an NFT trade or transfer of the Terraform NFT collection during the aforementioned time period.
  - Cycle 4 => common funder
  - Cycle 3, 5, 6 => transfer funds from earned wash trade
- Short simple cycles vs long directed cycles vs asset transfer graph

Method	Wash Trade Share (Dataset 1)	Wash Trade Share (Dataset 2)	FA's Wash Trade Share
SCC A	8.37	45.89	99.76
SCC AU	16.19	51.25	
SCC AC	9.68	64.41	

Wash Trade Share of Terraforms for all detection methods



Simple cycles with ETH transfers of max depth 2

Evaluation: Method SCC AU / AC

- **Removing addresses not found in both graph and FA's dataset**
- **Average Results:** 86.24% and 81.44% against 46% of SCC A or 64% of SCC AC

Method	Suspicious Wallet Addresses Flagged (%)	Suspicious Tx Flagged (%)
SCC A	77.98	85.10
SCC AU	95.03	89.23
SCC AC	71.33	84.40

Comparison against Footprint Analytics' data on suspicious addresses and transactions

Detecting and quantifying wash trading of non-fungible tokens

Conclusion

Conclusion and Improvements

- **Dataset**
  - **ETH Node:** Full data and data reliability through cryptographic checks, and faster than putting together data from varying sources
  - **Less Collections:** 20 NFT collections caused weeks of data scraping work to be done
  - **Flagged Addresses:** Increase data reliability by using industry standards of flagged addresses, like CEX's wallet address, etc
- **Intention detection:** Using price changes as a reliable metric
- **More eval metrics:** Evaluations could be better with more examples and thought into metrics

Detecting and quantifying wash trading of non-fungible tokens

Thank You